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Dear Editor-in-Chief,
Computational Optimization and Applications
Springer

We are pleased to submit our manuscript entitled ”From market states to robust portfolios: grid-restricted adaptive CVaR optimization with real financial data” for consideration for publication as a Research Article in Computational Optimization and Applications.

In this manuscript, we address a fundamental limitation in robust portfolio optimization: the reliance on either rigid, discrete market regimes or unconditional ambiguity sets. To bridge this gap, we propose a novel continuous-state robust portfolio optimization framework formulated as a Semi-Infinite Program (SIP). By defining a continuous, compact state space governed by observable financial indicators (CBOE VIX and equity market drawdown), our model optimizes Conditional Value-at-Risk (CVaR) across an infinite family of kernel-weighted empirical conditional return distributions.

To solve the resulting SIP, we implement an adaptive constraint generation algorithm over a finite grid-restricted separation oracle. The manuscript presents the theoretical properties of the continuous model and validates the finite approximation via grid-dispersion diagnostics. A comprehensive computational experiment, consisting of a 31-year rolling out-of-sample backtest, demonstrates the tractability of the constraint-generation scheme while documenting the empirical risk-reward tradeoffs associated with state-robust optimization.

This combination of continuous-state robust modeling, finite-grid constraint generation, and extensive computational evaluation relates directly to the scope of Computational Optimization and Applications.

In accordance with the journal’s ethical guidelines, we confirm that this manuscript is original, has not been published before, and is not currently under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to this journal.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Madani Bezoui (Corresponding Author)
CESI LINEACT, UR 7527, Nancy, France
Email: mbezoui@cesi.fr

Thiziri Sifaoui
Department of Mathematics and Computer Science, University of Amine Elokkal El Hadj

Moussa Eg Akhamouk, Tamanghasset, Algeria
LAROMAD, Faculty of Sciences, UMMTO, Tizi Ouzou, Algeria